

## Arithmetic sequences



1 Determine whether each of the following sequences is an arithmetic progression:

a  $1, \sqrt{5}, 5, 5\sqrt{5}, \dots$

b  $4, 7, 10, 13, \dots$

c  $2, 2^2, 2^4, 2^6, \dots$

d  $2, 0, -2, -4, \dots$

e  $3, -3, 3, -3, \dots$

f  $3, 6, 12, 24, \dots$

g  $-32, -26, -20, -14, \dots$

h  $5, 7, 5, 7, \dots$

2 How is the common difference of an arithmetic sequence obtained?

3 Find the common difference of the following arithmetic sequences:

a  $2, 0, -2, -4, \dots$

b  $7, 18, 29, \dots$

c  $330, 280, 230, 180, \dots$

d  $-6, -\frac{39}{7}, -\frac{36}{7}, -\frac{33}{7}, \dots$

4 Write down the next two terms of the following arithmetic sequences:

a  $4, 8, 12, \dots$

b  $6, 2, -2, -6, \dots$

c  $9, 14, 19, \dots$

d  $3.5, 5, 6.5, \dots$

5 Find the missing terms in the following arithmetic progressions:

a  $8, \_, 16, 20, \_$

b  $\_, 0, \_, 10, \_$

6 For each of the following, write the first four terms in the arithmetic progression.

a The first term is  $-10$  and the common difference is  $4$ .

b The first term is  $-8$  and the common difference is  $-2$ .

c The first term is  $a$  and the common difference is  $d$ .

7 Consider the arithmetic sequence  $1.4, 2.3, 3.2, \dots, 10.4$

a Determine  $d$ , the common difference.

b Find  $n$ , the number of terms in the sequence.

8 Insert five terms in the arithmetic sequence which has  $-12$  as its first term and  $24$  as its last term:

$$-12, \_, \_, \_, \_, \_, 24$$

- 9 Find the value of  $x$  such that  $x + 4$ ,  $6x + 5$ ,  $9x - 8$  form successive terms in an arithmetic progression.
- 10 An arithmetic progression has  $u_5 = 21$  and  $u_{19} = 77$ .
- Find the value of  $d$ .
  - Find the value of  $u_1$ .
  - Find  $u_{11}$ , the 11th term in the sequence.
- 11 An arithmetic progression has  $u_6 = -11$  and  $u_8 = -17$ .
- Find the value of  $d$ .
  - Find the value of  $u_1$ .
  - Hence find the 14th term in the progression.
- 12 In an arithmetic progression, the first term is 32. Given that the 9th term is 4 times the 5th term, find the common difference  $d$ .
- 13 The first three terms of an arithmetic progression are: 34, 27, 20, . . .
- Find the range of values of  $n$  for which the terms in the progression are positive.
  - Hence determine the number of positive terms in the progression.
  - Find the last positive term in the progression.
- 14 In an arithmetic progression the 6th term is  $x$  and the 10th term is  $y$ .
- Form an expression for  $d$  in terms of  $x$  and  $y$ .
  - Form an expression for  $a$  in terms of  $x$  and  $y$ .
  - Hence, find an expression for the 16th term in terms of  $x$  and  $y$ .
- 15 The 14th term of an arithmetic progression is equal to the sum of the 5th term and the 10th term. The common difference is  $-2$ .
- Find the value of  $u_1$ .
  - Hence, find the 9th term.
- 16 Consider the sequence 2, 7, 12, . . . . If the  $n$ th term is 132, find the value of  $n$ .
- 17 Consider the sequence 2,  $-3$ ,  $-8$ , . . . . If the  $n$ th term is  $-578$ , find the value of  $n$ .

## Explicit rules



**18** For each of the following sequences:

- i Determine  $d$ , the common difference.
- ii State the expression for finding  $u_n$ , the  $n$ th term in the sequence.
- iii Determine  $u_{10}$ , the 10th term in the sequence.

**a**  $5, 12, 19, \dots$       **b**  $17, 16.2, 15.4, \dots$       **c**  $10, 3, -4, \dots$       **d**  $5, \frac{23}{4}, \frac{13}{2}, \dots$

 **19** For each of the following definitions of the  $n$ th term of a sequence:

- i List the first four terms of the sequence.
- ii Find the common difference.

**a**  $u_n = 3n + 8$

**b**  $u_n = 11 + (n - 1) \times 10$

**c**  $u_n = -7 - 3(n - 1)$

 **20** For the each of the following formulas for the  $n$ th term of an arithmetic progression:

- i Find  $u_1$ , the first term of the sequence.
- ii Find  $d$ , the common difference.
- iii Find  $u_5$ , the 5th term in the sequence.

**a**  $u_n = 15 + 5(n - 1)$

**b**  $u_n = -8n + 28$

**c**  $u_n = 25 - 4(n - 1)$

**d**  $u_n = 12n - 9$

**21** Consider an arithmetic progression where  $u_7 = 44$  and  $u_{14} = 86$ .

- a** Determine  $d$ , the common difference.
- b** Determine  $u_1$ , the first term in the sequence.
- c** State the equation for  $u_n$ , the  $n$ th term in the sequence.
- d** Hence find  $u_{25}$ , the 25th term in the sequence.

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## Applications

**22** A termite treatment will cost \$250 for the first half hour, \$245 for the second half hour, \$240 for the third half hour and so on. Find the cost of a treatment that takes 6 hours.



- 23** A ball starts rolling down a slope. It rolls 25 cm during the first second, 53 cm during the second second, 81 cm during the third second and so on. At this rate, what is the total distance it will have rolled after 10 seconds?
- 24** A diving vessel descends below the surface of the water at a constant rate so that the depth of the vessel after 1 minute, 2 minutes and 3 minutes is 50 metres, 100 metres and 150 metres respectively.
- a** By how much is the depth increasing each minute?
  - b** What will the depth of the vessel be after 4 minutes?
  - c** Continuing at this rate, what will be the depth of the vessel after 10 minutes?
- 25** When a new school first opened,  $a$  students started at the school. Each year, the number of students increases by the same amount,  $d$ .  
At the beginning of its 7th year, it had 361 students. At the end of the 11th year, the school had 536 students.
- a** Find  $d$ , the number of students who joined the school each year.
  - b** Find  $u_1$ , the number of students that started at the school when it first opened.
  - c** At the end of the  $n$ th year, the school has reached its capacity at 921 students. Find the value of  $n$ .
- 26** A car bought at the beginning of 2009 is worth \$1500 at the beginning of 2015. The value of the car has depreciated by a constant amount of \$50 each year since it was purchased.
- a** What was the car purchased for in 2009?
  - b** Write an explicit rule for the value of the car after  $n$  years.
  - c** Find the year in which the car will be worth half the price it was bought for.